CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 25: Probability

Previous year board questions, topic-wise: 10 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Probability of an event

1. Which of these cannot be the probability of an event? (a) $\frac{5}{4}$ (b) 0.25 (c) $\frac{1}{33}$ (d) 67% ICSE 2025, 1 mark

2 Balls, numbered cards and letters

2. A box holds cards numbered 1 to 30, and one card is drawn at random. A wall clock is won for a perfect square, a water bottle for an even number that is also a multiple of 3, and a purse for a prime number. Find the probability of winning (a) the wall clock (b) the water bottle (c) the purse. ICSE 2025, 3 marks

3. The letters A, D, M, N, O, S, U, Y are written on separate cards. One card is drawn at random. Find the probability that the letter (a) is in the word MONDAY (b) is not in MONDAY (c) is in both SUNDAY and MONDAY. ICSE 2024, 3 marks

4. A letter is chosen at random from the English alphabet. Find the probability that it is a vowel. (a) $\frac{4}{26}$ (b) $\frac{5}{26}$ (c) $\frac{21}{26}$ (d) $\frac{5}{24}$ ICSE 2023, 1 mark

5. A bag holds 25 cards numbered 1 to 25, and one card is drawn at random. Find the probability that its number is (a) a multiple of 5 (b) a perfect square (c) a prime number. ICSE 2023, 3 marks

6. Each letter of the word AUTHORIZES is written on a separate disc, and one disc is drawn at random. Find the probability that the letter is (a) a vowel (b) one of the first 9 letters of the alphabet (c) one of the last 9 letters of the alphabet. ICSE 2020, 3 marks

7. A box holds 25 discs numbered 1 to 25, and one disc is drawn at random. Find the probability that its number is (a) odd (b) divisible by both 2 and 3 (c) less than 16. ICSE 2019, 3 marks

8. Cards numbered 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 are kept in a bag, and one is drawn at random. Find the probability that its number is (a) prime (b) divisible by 4 (c) a multiple of 6 (d) odd. ICSE 2018, 4 marks

9. Sixteen cards are labelled a, b, c, ..., o, p, and one is drawn at random. Find the probability that the letter is (a) a vowel (b) a consonant (c) not one of the letters of the word "median". ICSE 2017, 3 marks

3 Coins and dice

10. Assertion (A): When a die is thrown once, the probability of getting an even number is $\frac{2}{3}$. Reason (R): The even numbers on a die are $\{2, 4, 6\}$. (a) A true, R false (b) A false, R true (c) both true (d) both false ICSE 2024, 1 mark

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